Zhou Chen

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Education

Auburn University, Ph.D. Candidate in Computer Science and Software Engineering — Auburn, AL
Advisor: Sathyanarayanan N. Aakur — May 2024 - Present
- GPA: *4.0/4.0*

Auburn University, M.S. in Computer Science and Software Engineering — Auburn, AL
- GPA:*4.0/4.0* — Aug. 2022 - May. 2024

Central South University, B.S. in Computer Science and Technology — Changsha, China
- GPA: *3.3/4.0* — Sep. 2018 - Jun. 2021

Publications

EASE: Embodied Active Event Perception via Self-Supervised Free Energy Minimization
- **Zhou Chen**, Sanjoy Kundu, Harsimran Baweja, Sathyanarayanan N. Aakur
- *IEEE Robotics and Automation Letters (RA-L)*, 2025. Accepted. (Impact Factor: 5.3)

CRAFT: A Neuro-Symbolic Reasoning Framework for Visual Functional Affordance Grounding
- **Zhou Chen**, Joe Lin, Sathyanarayanan N. Aakur
- *19th Intl. Conf. on Neurosymbolic Learning and Reasoning (NeSy)*, 2025. Accepted.

Recursive Predictive Abstraction for Unsupervised Temporal Partonomy Induction
- **Zhou Chen**, Sathyanarayanan N. Aakur
- *Under review at ICLR 2027*

FUSE: Active Functional Affordance Grounding through Adaptive Semantic-Geometric Evidence Acquisition
- **Zhou Chen**, Sathyanarayanan N. Aakur
- *Under review at AAAI 2027*

Generalized Event Partonomy Inference with Structured Hierarchical Predictive Learning
- **Zhou Chen**, Joe Lin, Sathyanarayanan N. Aakur
- *Under review at AAAI 2027*

CRAFT-E: A Neuro-Symbolic Framework for Embodied Affordance Grounding
- **Zhou Chen**, Joe Lin, Sathyanarayanan N. Aakur
- *Under review at WACV 2027*

Prediction Method of Multiple Related Time Series Based on Generative Adversarial Networks
- Weijie Wu, Fang Huang, Yidi Kao, **Zhou Chen**, Qi Wu
- *Information*, Vol. 12, No. 2, Article 55, 2021. Accepted.

Class Geometry as Supervision for Sample-Efficient Open-World Detection
- Akash Rao, **Zhou Chen**, Revanth Reddy Palem, Udhav Ramachandran, Ruth Scimeca, Sathyanarayanan N. Aakur
- *Under review at AAAI 2027*

Workshops

Learning to Perceive and Act: Active Event Understanding via Predictive Free Energy Minimization
- Vision-based Assistants in the Real-World Workshop, CVPR 2025.

A Self-Supervised Framework for Embodied Active Event Perception
- *Presented at the How Do Robots Care? Workshop, ICRA 2025.*
- *Also presented in the Late-Breaking Results Track, ICRA 2025.*

Work Experience

Graduate Research Assistant May 2024 – Present
Open-ended Reasoning and Knowledge Acquisition (ORCA) Lab
Advisor: Dr. Sathyanarayanan N. Aakur

- Research in neuro-symbolic AI, affordance grounding, hierarchical event perception and embodied event perception.
- Developed EASE, a self-supervised hierarchical event perception model based on free-energy minimization for active visual understanding.
- Built and extended the CRAFT neuro-symbolic framework integrating commonsense knowledge and CLIP-based vision-language reasoning for interpretable and embodied affordance grounding in robotic manipulation.
- Designed generalized event partonomy inference for structured predictive learning across event hierarchies.

Hualu Zhida Technology Co., Ltd. Aug 2021 – Jun 2022
Junior Software Engineer, Embedded Department – Android Development Group Dalian, China

- Designed and deployed Android-based POS terminals used in Beijing's public transportation system, enabling real-time fare processing for thousands of daily users. Optimized system reliability and responsiveness under high concurrency and unstable network conditions.
- Led end-to-end development of intelligent driver sign-in devices with integrated modules for identity verification, alcohol testing, and schedule synchronization. Collaborated with backend and hardware teams to ensure robust device-software integration and compliance with safety regulations.

Internships

ZTE Corporation, China Sep 2020 – Jan 2021

- Developed Android applications with focus on UI module implementation and performance optimization.
- Collaborated in an agile team to debug and improve mobile app reliability.

Project

Task-Driven Exploration for Functional Affordance Grounding Jan. 2026 - Present

- Formulated the problem of task-driven exploration for functional affordance grounding: given a verb-level functional intent (e.g., cut, scoop, write), the agent must select a camera viewpoint where an object that affords the verb is both semantically grounded by CLIP/SAM and physically actionable for grasping.
- Implemented an explore-then-commit framework that couples CLIP/SAM-based semantic grounding with 3D Gaussian Splatting uncertainty, and benchmarked random, semantic-only, geometry-only, and task-conditioned policies.
- Designed evaluation metrics (Success@T, steps-to-success, regret, semantic/visibility coverage) to quantify exploration efficiency and final affordance grounding quality across varying occlusion and clutter levels.

B.S. Graduation Design (Thesis) May 2021

- **Topic**: Design of Experimental Software Tools for Time Series Forecasting Based on Deep Learning
- **Grade**: A
- **Objective**: Developed a visual experimentation platform for researching time series forecasting using a variety of deep learning and machine learning models, with applications in reinforcement learning scenarios.
- **Significance**: The tool provides an extensible and user-friendly environment for exploring time series prediction strategies, accelerating algorithm evaluation and facilitating research in time-dependent decision-making systems.

Professional Service

Reviewer:

- IEEE Robotics and Automation Letters (RA-L), 2025
- IEEE MultiMedia, 2026

Technical & Research Proficiencies

Programming Languages: C/C++, Java, Python
Frameworks & Tools: Proficient in PyTorch, ROS 2, Gym-UnrealCV
Research Interests: Active Event Perception, Robotics, Visual Affordance, Computer Vision, Learning-based Control